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Self-Supervised Domain Adaptation for Satellite Imagery

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Agenda

- Domain & Sub-Domain
- Introduction
- Literature Review
- Research Gaps
- Problem Statement
- Research Objectives
- Datasets
- Methodology
- Expected Outcomes
- Conclusion
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- **Domain:** Artificial Intelligence and Machine Learning (Computer Science)
- **Sub-domain:** Computer Vision (Remote Sensing / Satellite Image Analysis)
- **Specific focus area:** Domain Adaptation using Self-Supervised Learning and Generative Adversarial Networks (GANs)

Introduction

- Satellite AI models are trained on one sensor, season, or region – but deployed across many.
- Performance drops sharply under “domain shift,” even for identical land-cover classes.
- Standard fixes need labeled target-domain data and a clear domain boundary – both expensive, and often undefined in blended multi-sensor mosaics.

Literature Review

S.no	Title of Paper & Year	Author(s)	Methodology	Gaps
1.	Self Supervised Domain anaSS(DA) ² (Base Paper)	Zhang, Shi & Zhu (2023)	Self-supervised, domain-agnostic adaptation via a contrastive GAN; translates between any two satellite patches without labels	Built the method that works without needing label or a clear domain boundary, but it can compare only 2 satellites at a time.
2.	StarGAN: Unified GANs for Multi-Domain Image-to-Image Translation.	Choi et al. (2018)	A single generator handles translation across many domains at once, using a discrete domain label as conditioning	Built something that handles many domains at once but it needs someone to manually tag which domain each image belongs to.

S.no	Title of Paper & Year	Author(s)	Key Idea	Gaps
3.	StarGAN v2: Diverse Image Synthesis for Multiple Domains.	Choi et al. (2020)	Continuous style codes (via AdaIN) replace discrete labels, enabling diverse, non-deterministic synthesis	Improved on that, but was only tested on things like human faces and only check “does it look real” but not “does it actually helps the AI classify things correctly.
4.	StandardGAN: Multi-source Domain Adaptation for Semantic Segmentation of VHR Satellite Images by Data Standardization.	O Tasar et al. (2020)	Applies multi-source domain adaptation specifically to satellite semantic segmentation	Tested StarGAN on satellite images but struggling as it doesn’t handle unseen combinations and keeping the image content accurate.

Research Gaps

Nobody has built a method that does all 3 Things together:

- No labels needed.
- Works across many satellites.
- Checked by “Does it actually improve real results” instead of “Does it looks nice”.

Problem Statement

Deep learning models trained on satellite imagery from one sensor, season, or region degrade sharply when applied to another – and existing fixes require labeled target data and a pre-defined domain boundary, neither of which is practical at scale.

Therefore, we need ***Self-Supervised Multi-Domain Adaption for Satellite imagery***

Research Objectives

The objectives of this project are:

- To design and develop a domain encoder that learns continuous, continuous-valued domain embeddings directly from unlabeled multi-sensor satellite image patches.
- To develop a conditional generator that translates image content into the style of an arbitrary target domain using a single unified model.
- To analyze and evaluate model performance on sensor-pair combinations never jointly observed during training, in order to assess generalization capability.
- To evaluate the proposed method's downstream impact by measuring real classification/segmentation accuracy gains, rather than relying on visual realism metrics alone.

Datasets

1. MSC-France (Multi-Sensors Cities France)(Pretrain)

A large-scale, unlabeled dataset of $\approx 5,000$ image triplets covering French cities, pairing Sentinel-2, Landsat-8, and SPOT-6 imagery (with an optional high-resolution BDORTHO subset). Introduced in the X-STARS paper (arXiv:2405.09922). Used for self-supervised pretraining of the domain encoder, since its scale and lack of labels make it suited for learning domain embeddings without supervision.

2. PASTIS-HD (Train/Evaluate)

A labeled benchmark of 2,433 patches (128×128 px) across four agricultural regions in France, with Sentinel-1 (SAR), Sentinel-2 (optical), and SPOT 6-7 (very-high-resolution optical) co-registered at each location. Includes pixel-level annotations for 18 crop classes. Used as the primary dataset for training and evaluating the conditional generator, and for the downstream crop-classification accuracy benchmark.

3. MultiEarth 2022 (Train/Evaluate)

A four-sensor dataset (Sentinel-1, Sentinel-2, Landsat-5, Landsat-8) co-registered over the Amazon rainforest, with deforestation labels. Its four aligned sensors give six possible sensor-pair combinations, making it the only dataset able to support a genuine leave-one-pair-out generalization test. Also used as a second, independent downstream segmentation task.

4. SpaceNet 6 (Zero-shot test)

A dataset pairing Capella X-band SAR with WorldView-2 optical imagery over Rotterdam, with building-footprint labels. Reserved exclusively as a zero-shot generalization test — an unseen sensor over unseen geography — kept out of training entirely to avoid confounding sensor generalization with a change in location.

Datasets

Objectives	Datasets
Domain Encoder	MSC-France (primary) + PASTIS-HD unlabeled (secondary)
Conditional Generator	PASTIS-HD (primary) + MultiEarth 2022 (secondary)
Generalization Test	MultiEarth 2022 (only dataset that works)
Downstream Evaluation	PASTIS-HD + MultiEarth 2022 (secondary task)

Conclusion

- Extends SS(DA)²'s label-free adaptation from pairwise to a continuous multi-domain setting.
- Evaluated on downstream task accuracy – a stronger, more practical measure than visual realism alone
- Contributes toward lower-cost, more resilient satellite monitoring for disaster response and climate applications

Future Scope

- **Cross-Modality Adaptation** – Extend from optical–optical to optical–SAR translation.
- **Foundation Model Integration** – Combine with pretrained remote-sensing backbones such as SatMAE or Prithvi.
- **Change Detection** – Apply the framework to bi-temporal change-detection tasks, beyond static classification.
- **Adversarial Robustness** – Evaluate and harden the adaptation pipeline against adversarial perturbations.

References

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THANK YOU